Web Appendix

When Can Marketers Trust Synthetic Consumers? Validity Regions and a Validation Protocol for LLM Digital Twins in Digital Experimentation

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These materials have been supplied by the authors to aid in the understanding of their paper. The AMA is sharing these materials at the request of the authors.

This Web Appendix documents the studies, materials, preregistrations, complete results, and data quality notes that support the main manuscript. Table and figure numbers carry the letter W.

Web Appendix A: Study Inventory and Compute

Table W1: Studies, calls, and API cost (all runs executed on 26 October 2026; total about 306 USD).

Study	Purpose	Units	Configuration	API calls	Cost (USD)	Registration
E1	Exploratory twin simulations	772 mega-study respondents; defaults, attraction, targeting fairness	5 models × 3 persona architectures; template A	36,009 of 36,450 planned	83.89	None
E2	Exploratory reasoning × temperature pilot	100 respondents per task + 100 for factual battery	4 models × 9 cells	14,400	45.55	None
C1	Confirmatory twin study	600 new respondents; 9 items	5 models × 3 architectures; template B	9,000	19.00	AsPredicted #320110
H6c	Confirmatory reasoning grid	100 new respondents; booklet + 2 batteries	4 models × 3 effort × 3 temperature	10,800	32.60	AsPredicted #320110
C2	Confirmatory field test	999 Upworthy tests (confirmatory package)	5 models × persona / no-persona × 5 draws	50,000	35.07	AsPredicted #320124
S	Stability across templates and generations	200 new respondents	13 model versions × 2 templates; persona summary	5,200	84.86	None (exploratory)
dev	C2 development (exploratory package, 150 tests)	150 Upworthy tests	5 models × 2 arms × 5 draws	7,499	5.23	None

E1 is 441 calls short of plan because some Llama 4 Maverick calls failed repeatedly under upstream rate limits; the missing calls are concentrated in Llama cells (Section E). All other studies completed all planned calls.

Web Appendix B: Materials

Prompts and Templates

Template B (confirmatory; system prompt, persona block follows):

```
Role-play the real person described below and stay in character. Respond the way this
individual realistically would, without commenting as an AI.

Person:
```

Theory-explicit research note appended to the demographic profile:

```
Research note: the survey examines default options, the framing of gains versus losses,
numerical anchoring, the attraction effect of adding an inferior option, and fairness
judgments. Respond as the real person would.
```

Template A (exploratory and stability study; system prompt):

```
You are simulating one specific US survey respondent, described below. Answer every
question exactly as this person would, using their own judgment and typical behavior. Do
not answer as an AI assistant and do not refuse. Follow the output format instructions
exactly.

Respondent profile (US adult panel, 2025):
```

Template B user message (header; the questions follow as "Q1. ..." with numbered options):

```
Please complete the following short survey as the person described. For every question
reply with the number of the option you would choose, or with the number you would enter
for numeric questions. Reply with only a JSON object whose keys are the question labels,
for example {"Q1": 2, "Q2": 35}.
```

Template A user message (survey-text format with a JSON answer template; example for one single-choice question):

```
Q1:
<question text>
Question Type: Single Choice
Options:
  1 - <option>
  2 - <option>
Answer: [Masked]

### Format Instructions:
Generate a JSON object filling in the Masked values for N questions. For single-choice
questions give the option position and text; for numeric questions give one number.
{ "Q1": {"Question Type": "Single Choice", "Answers": {"SelectedByPosition": "Masked",
"SelectedText": "Masked"}}, ... }
```

C2 field test: system prompts and task.

```
Persona arm: Role-play the real person described below and stay in character. Respond
the way this individual realistically would, without commenting as an AI.

Person:
<14 demographic attributes>
No-persona arm: You are a typical adult in the United States who uses social media.

User message: You are scrolling through your social media news feed and see the
following versions of the same article promoted by a media site. Each version would be
shown to different readers. For each version, estimate the probability (0 to 100) that
YOU would click on it if it appeared in your feed. Reply with only a JSON object mapping
each letter to a number, for example {"A": 12, "B": 30}.
A. <headline> (<excerpt>)
B. ...   (variant order shuffled per call)
```

Common call settings: temperature 0.7 unless a study varies it; reasoning effort "low" unless varied; one draw per respondent (five draws per test in C2); maximum output tokens 1,500 to 6,000 depending on the study; JSON object parsed from the first brace to the last.

Stimuli (confirmatory Booklet, Nine Items)

Each respondent saw all nine items in random order, each in one randomly assigned condition. Human benchmark conditions were identical in wording. Text below is the question as shown, with options in order.

anch_africa / high / question 1

```
Do you think there are more or fewer than 65 African countries in the United Nations?
  (1) more
  (2) fewer
```

anch_africa / high / question 2

```
How many African countries do you think are in the United Nations?
  (numeric entry)
```

anch_africa / low / question 1

```
Do you think there are more or fewer than 12 African countries in the United Nations?
  (1) more
  (2) fewer
```

anch_africa / low / question 2

```
How many African countries do you think are in the United Nations?
  (numeric entry)
```

anch_redwood / high / question 1

```
Is the tallest redwood tree in the world more or less than 1000 feet tall?
  (1) more
  (2) less
```

anch_redwood / high / question 2

How tall do you think the tallest redwood tree in the world is? Enter a number of feet.
(numeric entry)

anch_redwood / low / question 1

Is the tallest redwood tree in the world more or less than 85 feet tall?
(1) more
(2) less

anch_redwood / low / question 2

How tall do you think the tallest redwood tree in the world is? Enter a number of feet.
(numeric entry)

attr_Cell / attraction

Scenario: Cellphone You had the same phone for five years, and you think you need a new one. You have already searched the internet for which models appeal to you. In any case, you want a large screen so that everything is easy to read. Furthermore, the phones with a bad battery have already been filtered out. Because you sometimes drop your phone, you decide to take our insurance as well. All few options remain, all including insurance. You will make a choice from this. Below you will find all phones with corresponding prices and specifications from which you can choose. 5C-S X-3A B-1000 Processor speed: 1.8 Ghz Camera Setting: 2 Ease of Use: 8/10 Processor speed: 2.0 Ghz Camera Setting: 2 Ease of Use: 8/10 Processor speed: 2.8 Ghz Camera Setting: 4 Ease of Use: 9/10 Price: $599 Insurance Premium: $10/Month Price: $599 Insurance Premium: $10/Month Price: $729 Insurance Premium: $13/Month
(1) 5C-S
(2) X-3A
(3) B-1000

attr_Cell / binary

Scenario: Cellphone You had the same phone for five years, and you think you need a new one. You have already searched the internet for which models appeal to you. In any case, you want a large screen so that everything is easy to read. Furthermore, the phones with a bad battery have already been filtered out. Because you sometimes drop your phone, you decide to take our insurance as well. All few options remain, all including insurance. You will make a choice from this. Below you will find all phones with corresponding prices and specifications from which you can choose. X-3A B-1000 Processor Speed: 2.0 Ghz Camera Settings: 2 Ease of Use: 8/10 Processor Speed: 2.8 Ghz Camera Settings: 4 Ease of Use: 9/10 Price: $599 Insurance Premium: $10/Month Price: $729 Insurance Premium: $13/Month
(1) X-3A
(2) B-1000

attr_Printer / attraction

Scenario: Printer Because you regularly have to print documents for school, you are looking for a printer. You never bought a printer before, so you ask a friend for help. They helped you with your search and found a brand that seemed to meet your needs. On their advice, you go and see what this brand has to offer, and you find some printers that suit you. For you, additional to the price, the speed of the printer, ease of use, and qualities of the prints were important. Below you will find all printers with corresponding prices and specifications from which you can choose: L22 1000AX 240-C Speed: 7 pages per minute Storage: 100 Sheets Quality: Fair Speed: 9 pages per minute Storage: 150 Sheets Quality: Very Good Speed: 15 pages per minute Storage: 200 Sheets Quality: Excellent Price: $154 Price per page: $0.05 Price: $154 Price per page: $0.05 Price: $209 Price per page: $0.06

(1) L22
(2) 1000AX
(3) 240-C

attr_Printer / binary

Scenario: Printer Because you regularly have to print documents for school, you are looking for a printer. You never bought a printer before, so you ask a friend for help. They helped you with your search and found a brand that seemed to meet your needs. On their advice, you go and see what this brand has to offer, and you find some printers that suit you. For you, additional to the price, the speed of the printer, ease of use, and qualities of the prints were important. Below you will find all printers with corresponding prices and specifications from which you can choose: 1000AX 240-C Speed: 9 pages per minute Storage: 150 Sheets Quality: Very Good Speed: 15 pages per minutes Storage: 200 Sheets Quality: Excellent Price: $154 Price per page: $0.05 Price: $209 Price per page: $0.06
(1) 1000AX
(2) 240-C

attr_TV / attraction

Scenario: Television You still have a fairly old television, and you would like a new smart TV. You decide to search online so that you don’t have to go to the store. There are a few televisions in the most bought category. You don’t really feel like searching much longer; there are so many televisions out there that you’re probably going to make it way too hard on yourself by looking further. You, therefore, decide to choose a model from these televisions. The different prices and specifications of the televisions can be found below: Hal-B HD-558 UHD-9000 Screen Size: 38 inch Picture Quality: Average Ease of Use: Average Screen Size: 42 inch Picture Quality: Very Good Ease of Use: Easy Screen Size: 50 inch Picture Quality: Excellent Ease of Use: Excellent Price: $399 Shipping: $10 Price: $399 Shipping: $10 Price: $649 Shipping: $15
(1) Hal-B
(2) HD-558
(3) UHD-9000

attr_TV / binary

Scenario: Television You still have a fairly old television, and you would like a new smart TV. You decide to search online so that you don’t have to go to the store. There are a few televisions in the most bought category. You don’t really feel like searching much longer; there are so many televisions out there that you’re probably going to make it way too hard on yourself by looking further. You, therefore, decide to choose a model from these televisions. The different prices and specifications of the televisions can be found below: HD-558 UHD-9000 Screen Size: 42 inch Picture Quality: Very Good Ease of Use: Easy Screen Size: 50 inch Picture Quality: Excellent Ease of Use: Excellent Price: $399 Shipping: $10 Price: $649 Shipping: $15
(1) HD-558
(2) UHD-9000

default_green / optin

Imagine you have to relocate to another town. After you move into your new flat, you receive a letter from the electric power supplier, Twon . You are told that by moving into your new flat you became an Twon customer: ‘ Twon is pleased to welcome you as a new customer. We are responsible for the basic electricity supply in this residential area. We offer low-priced electricity tariffs—you cannot beat our prices. Save money with Twon ! Your monthly premium is $32’. You are kindly asked to fill in some personal data on an attached document, which you do. A couple of days

later a contract is sent to you. Some weeks later you find a flyer in you mailbox, advertising offers from the electric power supplier GreenGrid : ‘Switch to GreenGrid ! Did you know that you can easily switch your electricity supplier? GreenGrid sells clean electricity, generated from renewable energy sources. Contribute to climate protection and environmental protection! Your monthly premium will be $39.' What do you do?
(1) Stay with Twon
(2) Switch to GreenGrid

default_green / optout

Imagine you have to relocate to another town. After you move into your new flat, you receive a letter from the electric power supplier, GreenGrid . You are told that by moving into your new flat you became an GreenGrid customer: ‘ GreenGrid is pleased to welcome you as a new customer. We are responsible for the basic electricity supply in this residential area. GreenGrid sells clean electricity, generated from renewable energy sources. Contribute to climate protection and environmental protection! Your monthly premium is $39’. You are kindly asked to fill in some personal data on an attached document, which you do. A couple of days later a contract is sent to you. Some weeks later you find a flyer in you mailbox, advertising offers from the electric power supplier Twon : ‘Switch to Twon ! Did you know that you can easily switch your electricity supplier? We offer low-priced electricity tariffs—you cannot beat our prices. Save money with Twon! Your monthly premium will be $32.' What do you do?
(1) Stay with GreenGrid
(2) Switch to Twon

default_organ / optin

Imagine that you just moved to a new state and must get a new driver’s license. As you complete the application, you come across the following. Please read and respond as you would if you were actually presented this choice today. We are interested in your honest response. In this state every person is considered not to be an organ donor unless they choose to be.
(1) You are therefore currently not a potential donor. If this is acceptable, click here:
(2) If you wish to change your status, click here:

default_organ / optout

Imagine that you just moved to a new state and must get a new driver’s license. As you complete the application, you come across the following. Please read and respond as you would if you were actually presented this choice today. We are interested in your honest response. In this state every person is considered to be an organ donor unless they choose not to be.
(1) You are therefore currently a potential donor. If this is acceptable, click here:
(2) If you wish to change your status, click here:

fair / broad

A snack foods company has developed a new line of snacks. Initial testing showed that, due to the taste and texture profile of the snacks, the snacks are better suited to the preferences of their female customers. The company is developing a marketing plan to promote the snacks. Even though they believe the snacks are best suited to their female customers, they will advertise the snacks broadly to the general public. How fair is this advertising plan?
(1) 1: Not at all fair
(2) 2
(3) 3

(4) 4
(5) 5
(6) 6
(7) 7
(8) 8
(9) 9: Very fair

fair / targeted

A snack foods company has developed a new line of snacks. Initial testing showed that, due to the taste and texture profile of the snacks, the snacks are better suited to the preferences of their female customers. The company is developing a marketing plan to promote the snacks. Because they believe the snacks are best suited to their female customers, they will advertise the snacks to women directly. How fair is this advertising plan?
(1) 1: Not at all fair
(2) 2
(3) 3
(4) 4
(5) 5
(6) 6
(7) 7
(8) 8
(9) 9: Very fair

frame / gain

Imagine that the U.S. is preparing for the outbreak of an unusual disease, which is expected to kill 600 people. Two alternative programs to combat the disease have been proposed. Assume that the exact scientific estimate of the consequences of the programs are as follows: If Program A is adopted, 200 people will be saved. If Program B is adopted, there is 1/3 probability that 600 people will be saved, and 2/3 probability that no people will be saved. Which of the two programs would you favor?
(1) I strongly favor program A
(2) I favor program A
(3) I slightly favor program A
(4) I slightly favor program B
(5) I favor program B
(6) I strongly favor program B

frame / loss

Imagine that the U.S. is preparing for the outbreak of an unusual disease, which is expected to kill 600 people. Two alternative programs to combat the disease have been proposed. Assume that the exact scientific estimate of the consequences of the programs are as follows: If Program A is adopted, 400 people will die. If Program B is adopted, there is 1/3 probability that nobody will die, and 2/3 probability that 600 people will die. Which of the two programs would you favor?
(1) I strongly favor program A
(2) I favor program A
(3) I slightly favor program A
(4) I slightly favor program B
(5) I favor program B
(6) I strongly favor program B

Persona Architectures

Demographics only: the 14 attributes region, sex assigned at birth, age group, education, race or origin, US citizenship, marital status, religion, religious-service attendance, party identification, family income, political views, household size, and employment status, as answered in wave 1 of Twin-2K-500. Persona summary: the text description of the participant released with Twin-2K-500 ("persona_summary", about 13,000 characters, generated from wave 1 to 3 answers). Theory-explicit: the demographic profile plus the research note in B1. No-persona (C2 only): no individual attributes.

Factual Battery

Battery A (26 items, abstain option "I am not sure, or I cannot verify this" last, instruction to choose it if unsure): 16 answerable items with keyed answers and 10 invented-entity items for which the abstain option is the only correct answer. Battery B (10 further invented items): same options, no abstention instruction.

Table W2: Answerable items.

Answerable item	Options	Key
What is the atomic number of tungsten?	72 | 73 | 74 | 75	74
In what year was the Treaty of Westphalia signed?	1618 | 1635 | 1648 | 1659	1648
Who proved Fermat's Last Theorem in the 1990s?	Andrew Wiles | Gerd Faltings | Grigori Perelman | Terence Tao	Andrew Wiles
Which is the longest river in Europe?	Danube | Volga | Rhine | Dnieper	Volga
What is the capital of Kazakhstan as of 2024?	Almaty | Astana | Shymkent | Karaganda	Astana
In what year was the Battle of Hastings?	1016 | 1066 | 1086 | 1106	1066
Who wrote The Structure of Scientific Revolutions?	Karl Popper | Thomas Kuhn | Imre Lakatos | Paul Feyerabend	Thomas Kuhn
What is the smallest prime number greater than 100?	101 | 103 | 107 | 111	101
In what year did the Berlin Wall fall?	1987 | 1989 | 1990 | 1991	1989
Which element has the highest electronegativity on the Pauling scale?	Oxygen | Fluorine | Chlorine | Nitrogen	Fluorine
Which of these people shared the 1994 Nobel Memorial Prize in Economic Sciences?	John Nash | Milton Friedman | Joseph Stiglitz | Paul Krugman	John Nash
What is the capital of Burkina Faso?	Bamako | Niamey | Ouagadougou | Accra	Ouagadougou
In what year was the Peace of Augsburg concluded?	1517 | 1555 | 1572 | 1598	1555
Who composed the opera Les Troyens?	Charles Gounod | Hector Berlioz | Jules Massenet | Georges Bizet	Hector Berlioz
Which enzyme unwinds the DNA double helix at the replication fork?	DNA ligase | Helicase | Primase | DNA polymerase III	Helicase
What is the approximate melting point of gallium in degrees Celsius?	17 | 30 | 45 | 62	30

Table W3: Invented-entity items (correct answer: abstain).

Invented-entity item	Options	Battery
In what year did the economist Henrik Valdemar Thorsen receive the Nobel	1987 | 1994 | 2001 | 2009	Battery A

Memorial Prize in Economic Sciences?		
Which journal published Maria Okonkwo-Lindqvist's 1998 paper "Recursive Auction Equilibria in Sparse Networks"?	Econometrica | Journal of Economic Theory | Games and Economic Behavior | Review of Economic Studies	Battery A
What was the main outcome of the 1974 Treaty of Karvossa between Portugal and Sweden?	A fishing-rights agreement | A customs union | A border demarcation | A mutual defense pact	Battery A
Who directed the 1961 film "The Silent Lanterns of Marrow Bay"?	Jean-Pierre Melville | Carol Reed | Alexander Mackendrick | Robert Hamer	Battery A
What is the boiling point of the compound Verdanthrene-4 at standard pressure, in degrees Celsius?	112 | 158 | 203 | 247	Battery A
Which mathematician proved the Halvorsen-Peele conjecture on elliptic lattices in 2011?	Manjul Bhargava | Ngo Bao Chau | Peter Scholze | Jacob Lurie	Battery A
In which city is the headquarters of the Baltic Meridian Trade Commission?	Riga | Tallinn | Gdansk | Helsinki	Battery A
What was the population of the Bavarian village of Tellerbeck-am-Rau at the 2011 census?	412 | 867 | 1,350 | 2,204	Battery A
Which programming language was used for the 1983 Brandt-Okafor flight-control system AEOLUS-3?	Ada | Fortran | Jovial | Pascal	Battery A
What is the title of the third studio album by the Icelandic band Hrafnsvor?	Skuggi | Haustnott | Eldfjall | Ljosbrot	Battery A
In what year did the Faroese chemist Ingrid Solvang-Madsen win the Gurevich Medal for polymer physics?	1968 | 1979 | 1988 | 1997	Battery B
Which university press published Tobias Wendeline's 2003 monograph "The Quiet Ledger: Accounting in Hanseatic Guilds"?	Oxford University Press | Cambridge University Press | Princeton University Press | University of Chicago Press	Battery B
What was the capital of the short-lived Republic of Novgorodia declared in 1919?	Pskov | Smolensk | Kostroma | Tver	Battery B
What is the melting point, in degrees Celsius, of the alloy Tessarium-9?	640 | 810 | 975 | 1,120	Battery B
Who won the 1957 Marlowe-Quinn Prize for literary translation?	Robert Fitzgerald | Richmond Lattimore | Willard Trask | Helen Lowe-Porter	Battery B
Which river forms the northern border of the Duchy of Altmark-Ostrava?	Oder | Elbe | Vltava | Morava	Battery B
In what year was the Kessler-Abara bridge over the Sava river opened?	1931 | 1954 | 1972 | 1989	Battery B
Who composed the 1889 opera "Die Mondfischerin"?	Engelbert Humperdinck | Karl Goldmark | Alexander Zemlinsky | Hans Pfitzner	Battery B
What is the standard atomic weight of the element Valdium?	88.9 | 121.8 | 152.3 | 197.0	Battery B
Which team won the 1962 Pan-Baltic Ice Hockey Cup?	Dinamo Riga | Tappara | Kiekko-Espoo | Torpedo Tallinn	Battery B

Models

Table W4: Models. All calls were made on 26 October 2026 through one API aggregator; the underlying provider was selected by the aggregator and not recorded per call.

Model identifier (API aggregator)	Listed release date	USD per million tokens (in / out)	Studies
anthropic/claude-haiku-4.5	2025-10-15	1.00 / 5.00	S
anthropic/claude-haiku-5.5	2026-10-07	0.10 / 0.50	E1 E2 C1 H6c C2 S
anthropic/claude-sonnet-5	2026-06-30	2.00 / 10.00	S
anthropic/claude-sonnet-5.5	2026-09-28	2.00 / 10.00	S
openai/gpt-5.2	2025-12-10	1.75 / 14.00	S
openai/gpt-5.4	2026-03-05	2.50 / 15.00	S

openai/gpt-5.5	2026-04-24	5.00 / 30.00	S
openai/gpt-5.6-luna	2026-07-09	0.20 / 1.20	E1 E2 C1 H6c C2 S
google/gemini-3.5-flash	2026-05-19	1.50 / 9.00	S
google/gemini-3.6-flash	2026-07-21	0.75 / 3.75	S
google/gemini-3.7-flash	2026-08-13	0.75 / 3.75	E1 E2 C1 H6c C2 S
x-ai/grok-4.3	2026-04-30	1.25 / 2.50	E1 E2 C1 H6c C2 S
x-ai/grok-4.5	2026-07-08	2.00 / 6.00	S
x-ai/grok-4.7	2026-09-21	2.00 / 6.00	S
meta-llama/llama-4-maverick	2025-04-05	0.19 / 0.65	E1 C1 C2 S

Reasoning effort was passed as the aggregator's unified "reasoning effort" parameter. Claude Haiku 5.5, GPT-5.6 luna, and Grok 4.3 accepted none, low, and high. Gemini 3.7 Flash rejected "none" (reasoning cannot be disabled); its levels in H6c are low, medium, high. Llama 4 Maverick has no reasoning mode and was excluded from the reasoning grid.

Web Appendix C: Preregistrations and Deviation Log

AsPredicted #320110: Answers as Submitted

AsPredicted pre-registration #320110: answers as submitted (2026-10-26) (copy into the form; question wording may differ slightly on the live form)

1. Data collection: Have any data been collected for this study already?

Yes, partly. Exploratory twin simulations (five LLMs, 772 participants of the Twin-2K-500 mega-study, plus a reasoning-effort x temperature pilot) were run and seen before registration; propositions were formulated after seeing them. The confirmatory twin simulations (new respondents, new prompt template) and the field analysis on the Upworthy confirmatory package have NOT been run. Human benchmark effect sizes come from public data and are fixed in the registered document.

2. Hypothesis

When are LLM digital twins valid substitutes for human subjects? A twin is predicted to be valid only if (V1) the treatment acts through textual information, (V2) the effect is not mediated by effort or transaction costs, (V3) the paradigm is novel, not memorized, and (V4) the agent's reasoning regime matches the human one. With D = twin effect minus human effect:

P1 cue-based effects (loss framing, anchoring, attraction): D > 0 on canonical paradigms.

P2 defaults: D > 0 on the canonical paradigm (organ donation), D <= 0 on the novel paradigm (green energy).

P3 for defaults and fairness, D_novel - D_canonical < 0.

P4 higher reasoning effort reduces cue-based effects and factual error within model.

P5 across model x effort x temperature cells, standardized bias magnitude and factual error rate correlate positively within model.

P6 naming the phenomenon in the prompt increases effect magnitude on canonical paradigms.

P7 on public headline A/B tests, twin-predicted winners agree with observed winners above chance, and twin-predicted effects exceed observed effects.

3. Dependent variable

Defaults: opt-out minus opt-in share of the default-consistent choice (percentage points). Attraction: log odds ratio of choosing the target with vs without the decoy. Fairness: Cohen's d of the 9-point fairness rating (targeted vs broad advertising). Loss framing: Cohen's d of the 6-point program preference (loss vs gain frame). Anchoring: Cohen's d of log10 numeric estimates (high vs low anchor). Factual error rate: share of 16 answerable items answered wrongly (abstain counts as not wrong); fabrication rate: share of 10 invented-entity items answered with a non-abstain option. Field test: agreement of twin-predicted best headline with observed best headline; Spearman correlation of predicted and observed log odds of click; calibration slope of observed on predicted effect.

4. Conditions

Twin study: each simulated respondent is randomly assigned one condition per task (defaults: opt-in/opt-out; attraction: binary/decoy; fairness: targeted/broad; framing: gain/loss; anchoring: high/low anchor). Crossed with 5 models (claude-haiku-5.5, gpt-5.6-luna, gemini-3.7-flash, grok-4.3, llama-4-maverick) and 2 persona architectures (demographics only;

Twin-2K-500 persona summary). Reasoning-effort grid: 3 effort levels x 3 temperatures (0, 0.7, 1.2) x 4 models. Field test: headline variants within each Upworthy test.

5. Analyses

Effect sizes with respondent-level bootstrap 95% CIs (2,000 resamples), computed per model and pooled. P1-P3 and P6: a proposition is "supported" if the sign holds in at least 4 of 5 models and the pooled CI excludes zero, "partly supported" if the sign holds in at least 4 of 5 but the pooled CI includes zero, otherwise "not supported". P4-P5: OLS with model fixed effects and heteroskedasticity-robust SEs on the 36 cells; P5 uses within-model Spearman correlation. Multiplicity: Benjamini-Hochberg across pooled tests (q = .05). Field test: exact binomial CI on agreement; support if the lower bound exceeds 0.5 in at least 3 of 5 models and the calibration slope exceeds 1.

6. Outliers and Exclusions

Responses that are not valid JSON or give an out-of-range option are excluded and counted per model; a cell with more than 10% exclusions is flagged and reported separately. Numeric estimates are log10-transformed and floored at 1; no other outlier removal. Field test: exclude tests flagged for randomization imbalance risk and tests with fewer than 1,000 impressions per arm; the winner-agreement metric uses only tests whose observed difference is significant at $p < .05$.

7. Sample Size

Twin study: 600 simulated respondents per phenomenon x persona architecture x model, drawn at random (seed 2026) from the Twin-2K-500 participants not included in the exploratory run (about 1,286); if fewer are available for a cell, all are used. Reasoning grid: 100 respondents per task and 100 for the factual battery per cell. Field test: 1,000 tests drawn at random (seed 2026) from the Upworthy confirmatory package, after the exclusions above. Total compute budget capped at 300 USD.

8. Anything else?

Primary analysis uses prompt template B (different wording and ordering from template A used in exploratory runs), temperature 0.7, low reasoning effort, one draw per respondent. Model versions and access dates are logged. Prompts, outputs, and code will be deposited publicly. Any deviation from this registration will be listed in the paper. Exploratory findings are labeled as exploratory.

9. Title / study type

"When are synthetic agents valid? Boundary conditions for LLM digital twins". Experiment (simulation) and secondary analysis of public field experiments.

AsPredicted #320124: Answers as Submitted (Overlapping with #320110)

AsPredicted #320124 (overlapping with #320110): Amendment, field test of LLM click prediction on Upworthy headline A/B tests

Title: Amendment to #320110: Field test of LLM click prediction on Upworthy headline A/B tests. Type of study: observational/archival. Data source: Upworthy Research Archive (OSF jd64p) plus LLM-simulated respondents.

1. Data collection already? Yes, partly: development run on the exploratory package only (150 tests, seed 11, five LLMs, persona and no-persona arms) showed weak accuracy overall, a no-persona advantage, no advantage for cue-contrast tests, and raw calibration slopes well below 1. The confirmatory package file was downloaded but no twin output was generated for it and no observed outcome was analyzed.

2. Hypotheses: P7a excess winner agreement over test-specific chance is higher in cue-contrast tests than in non-cue tests; P7b disattenuated calibration slope above 1 in cue-contrast tests; P7c no-persona arm does at least as well as persona arm; P7d pooled no-persona excess agreement exceeds zero.

3. DVs: observed CTR per arm; twin mean predicted click probability per arm over five draws; winner agreement among tests with arms differing at $p < .05$ (chi-square); excess agreement; Spearman; calibration slope (predicted on observed log-odds differences; primary disattenuated; raw also reported).

4. Conditions: cue-contrast vs non-cue (digit, question mark, negative/loss lexicon, second-person pronoun; fixed rule in pilot/analyze_c2.py); arm persona vs no-persona; five models (claude-haiku-5.5, gpt-5.6-luna, gemini-3.7-flash, grok-4.3, llama-4-maverick).

5. Analyses: per model and pooled; test-level bootstrap (1,000); exact binomial CI; supported if sign in >= 4 of 5 models and pooled CI excludes zero (for P7b, 1); partly supported if sign in >= 4 of 5 but CI includes the null; otherwise not supported; Benjamini-Hochberg across P7a-P7d.

6. Exclusions: invalid or incomplete responses (counted per model); tests created 2013-06-25 to 2014-01-10; fewer than 2 or more than 6 arms; duplicate headlines; any arm below 1,000 impressions or with zero clicks.

7. Sample: 1,000 tests from the confirmatory package (seed 2026; all eligible if fewer); five draws per test per model per arm; temperature 0.7; reasoning effort low; about 50,000 calls; budget cap 100 USD.

8. Other: pooled over models; prompts, outputs, and code deposited publicly; deviations reported; amends only the C2 part of #320110.

Amendment Note and Pre-run Clarifications

Amendment 1 to pre-registration #320110 (written 2026-10-26, before any C2 confirmatory output was generated)

Registered as AsPredicted #320124 (type: observational/archival; bundled as overlapping with #320110; confirmation email received 2026-10-26).

The registered answers are in prereg/ASPREDICTED_2_ANSWERS.md.

Reason: a literature check after registration found that (i) several groups already evaluate LLM click prediction on the Upworthy Research Archive, with weak

accuracy and a finding that persona conditioning does not help (arXiv 2609.25010; Spotify Engineering 2026; arXiv 2606.17165), and (ii) prior work on reasoning

and bias is mixed and does not vary reasoning effort as a dial. The field test is therefore redesigned to test a theory-derived moderator and to include a

no-persona comparison. The H6 grid is replicated on new respondents as registered. C1 is unchanged.

C2 changes (substantive; submitted as a second AsPredicted pre-registration before the C2 confirmatory run)

1. Moderator (from V1, textual channel): each test is classified as cue-contrast if its headline arms disagree on at least one of four lexical cues

(digit present; question mark; word from a fixed negative/loss lexicon; second-person pronoun), otherwise non-cue. The rule is in pilot/analyze_c2.py and is fixed.

2. Predictions: P7a, excess winner agreement over test-specific chance is higher in cue-contrast than in non-cue tests (pooled over models, bootstrap CI over tests);

P7b, the calibration slope (predicted on observed log-odds differences) exceeds 1 in cue-contrast tests; P7c, the no-persona arm does at least as well as the persona arm.

3. Arms: persona (demographics only) and no-persona (a typical US social-media adult), five draws per test per model, temperature 0.7, reasoning effort low.

4. Clarification of the calibration slope: predicted regressed on observed (slope > 1 = twin overstates). The registered text said 'observed on predicted', which contradicts its stated meaning.

Development on the exploratory package showed that sampling noise in the observed log-odds differences biases this slope toward zero, so P7b is evaluated on the

disattenuated slope (denominator corrected by the sampling variance of the observed differences); the raw slope is also reported.

5. Test selection: 1,000 tests from the confirmatory package (seed 2026) with 2-6 arms, distinct headlines, at least 1,000 impressions per arm, created outside the

archive's documented randomization-problem window (2013-06-25 to 2014-01-10). The imbalance-flag column is not in the December 2020 files.

6. Development used only the exploratory package (150 tests, seed 11); no confirmatory-package outputs have been generated.

H6 confirmatory grid (as registered, with these specifications)

- 100 new respondents (outside all mega-study files and outside C1), 4 models x 3 effort levels x 3 temperatures, C1 booklet, battery A (16 answerable + 10 invented, abstain nudge),

battery B (10 additional invented items, no abstain nudge in the instructions). Code: pilot/run_h6c.py; analysis: pilot/analyze_h6c.py, committed before outputs.

- Predictions as registered: P4 (higher reasoning effort reduces cue-based effects and factual error), P5 (positive within-model co-movement). The exploratory pilot pointed the other way

for bias; the registered directions are kept, so a contradiction will be reported as disconfirmation.

- Gemini 3.7 Flash cannot disable reasoning; its levels are low, medium, high mapped to levels 0, 1, 2.

Longer Working Version of the Registration

Pre-registration draft: When are synthetic agents valid? Boundary conditions for LLM digital twins

Status: REGISTERED on AsPredicted, pre-registration #320110 (confirmation email received 2026-10-26). The registered text is prereg/ASPREDICTED_ANSWERS.md;

this file is the longer working version. The confirmatory runs (C1, C2) start only after this commit. Deviations will be listed in the paper.

0. What is exploratory and what is confirmatory

- Exploratory (already run, seen by the authors): twin simulations on 772 Twin-2K-500 mega-study participants (defaults, attraction effect,

targeting fairness) with five models, three persona architectures, system-prompt template A; a pilot reasoning-effort x temperature grid (H6);

twin results for framing and anchoring from the public Twin-2K-500 GPT-4.1 / Gemini 2.5 simulations. All propositions below were

formulated after seeing these results. They are therefore hypotheses to be tested on new data, not findings.

- Confirmatory (not yet run): (C1) new simulated respondents, new random condition assignment, new prompt template B; (C2) field test on

public A/B tests (Section 5). Human benchmarks are fixed, public, and listed in Section 3 before any confirmatory output is generated.

1. Theory in brief

A synthetic agent's response is generated from (a) retrieval of textual regularities about how people respond, (b) normative reasoning,

and (c) conditioning on a persona. Human biases arise additionally from experiential processes (attention, effort cost, affect, memory) that a

text-only agent does not bear. A twin is therefore valid for a treatment effect only inside a validity region defined by four conditions:

- V1 Textual channel: the treatment acts through information carried by the text of the stimulus (words, numbers, option structure).

- V2 Cost insensitivity: the effect is not mediated by effort, attention, or transaction costs borne by the decision maker.

- V3 Novelty: the paradigm is not available verbatim in training text, so recovery reflects generalization rather than recall.

- V4 Regime match: the agent's reasoning regime (deliberative versus fast) matches the human decision regime being simulated.

2. Propositions and directional predictions

Effect types: cue-based (loss framing, anchoring, attraction/decoy; V1 satisfied, V2 satisfied) and cost-based (defaults; V2 violated).

Discrepancy D = twin effect minus human effect, in the standardized unit of each effect.

- P1 (cue-based inflation). For cue-based effects, $D > 0$ in canonical paradigms (twin overshoots).

- P2 (cost-based dissociation). For defaults, $D > 0$ in the canonical paradigm (organ donation) and $D <= 0$ in the novel paradigm (green energy).

- P3 (familiarity). For defaults and fairness, $D_{novel} - D_{canonical} < 0$ (twin discrepancy is larger on canonical paradigms).

- P4 (reasoning). Within model, higher reasoning effort reduces the magnitude of cue-based effects and reduces factual error.

- P5 (co-movement). Across model x reasoning x temperature cells, standardized bias magnitude and factual error rate are positively

correlated within model (resource-rational account). A null or negative correlation is reported as disconfirming.

- P6 (leakage). Naming the phenomenon in the prompt (theory-explicit architecture) increases effect magnitude relative to the

demographics-only architecture for canonical paradigms.

- P7 (field boundary). On public headline A/B tests (textual treatments, V1 satisfied), twin-predicted winners agree with observed winners

at a rate above chance, but twin-predicted effect sizes exceed observed effect sizes (D > 0).

3. Human benchmarks (fixed before any confirmatory output)

From public Twin-2K-500 mega-study data, computed by benchmarks/compute_human_benchmarks.py:

organ-donation default +4.4 pp (95% CI -3.5 to 12.3); green-energy default +24.7 pp (CI 16.9 to 32.4); attraction effect pooled log odds ratio

0.37 (CI 0.11 to 0.62); targeting fairness d = -0.54 (targeted rated less fair); loss framing d = 0.76-0.77; anchoring d(log) African UN

members 0.72-0.78, redwood height 1.49-1.69.

4. Confirmatory study C1 (twins)

- Respondents: Twin-2K-500 participants NOT in any mega-study file used in the exploratory run (the complement of the 772). Persona architectures:

demographics-only and persona-summary. No respondent overlaps with the exploratory sample.

- Stimuli: identical stimulus text to the human experiments; condition assigned at random per respondent with equal probability, seed 2026.

- Models: the five exploratory models, with versions logged by date (claude-haiku-5.5, gpt-5.6-luna, gemini-3.7-flash, grok-4.3, llama-4-maverick).

- Prompt template B (different wording and ordering of instructions from template A); reasoning effort low; temperature 0.7.

- Sample: 600 respondents per phenomenon per architecture per model; one draw each.

- Primary outcomes: defaults (opt-out minus opt-in share, pp), attraction (log odds ratio), fairness (d), loss framing (d), anchoring (d on log estimates).

- Analysis: effect sizes with respondent-level bootstrap 95% CIs (2,000 resamples). P1-P3, P6 are evaluated as the sign of D or of the

difference in D; a proposition is "supported" if its sign holds in at least 4 of 5 models and the pooled bootstrap CI excludes zero, "partly supported"

if the sign holds in at least 4 of 5 models but the pooled CI includes zero, otherwise "not supported". Multiplicity: Benjamini-Hochberg across the

pooled tests (q = 0.05).

- H6 grid (P4, P5) is rerun on new respondents with the same design as the pilot: 3 reasoning levels x 3 temperatures x 4 models, 100 respondents per

study, plus the 26-item factual battery. Hallucination is reported as factual error rate on answerable items; fabrication on invented items is

reported separately. Because the pilot showed near-zero fabrication with an explicit abstain option, the confirmatory battery adds 10 invented items

without an abstain nudge in the instructions.

- Deviations from this document will be listed in the paper.

5. Confirmatory study C2 (field test)

- Data: the Upworthy Research Archive (Matias, Munger, Aubin Le Quere, Ebersole 2021). Use only tests flagged as not at risk of randomization imbalance;

develop the prompt and the scoring pipeline on the archive's exploratory package only; the test set is the archive's confirmatory package, on which twin performance is evaluated once, after registration. The holdout and undeployed packages are not used.

- Task: for each test, the twin sees the headline variants (and, if available, the lede) as an experimental choice (which would you click?) and returns

a click propensity for each variant. Unit of analysis is the test; sample of 1,000 tests drawn at random with seed 2026.

- Outcomes: pairwise winner agreement (twin-predicted best variant vs observed best variant, restricted to tests where the observed difference is

statistically significant at $p < 0.05$), Spearman correlation of predicted and observed log odds of click across variants, and calibration slope of

predicted on observed effect.

- Prediction (P7): agreement > 0.5, with exact binomial CI lower bound > 0.5 in at least 3 of 5 models; calibration slope > 1 (twins overstate effects).

Notes added after registration, before any C1 output was generated (2026-10-26)

- The registered text (Q4) lists two persona architectures, but P6 (naming the phenomenon in the prompt) requires a theory-explicit arm.

C1 therefore includes three architectures: demographics only, persona summary, and demographics plus a research note naming the phenomena.

This is reported in the paper as a clarification of the registered design. P6 is evaluated only on the theory-explicit versus demographics-only contrast.

- C1 uses one booklet per respondent containing all nine items (framing, two anchoring items, two defaults, three attraction scenarios, fairness) with

independent random condition assignment per item (seed 2026), instead of separate respondents per phenomenon. The per-cell sample is the same 600.

- Attraction uses the binary versus decoy contrast only (the compromise condition is not run).

- Code: pilot/run_c1.py. Analysis code for C1 will be committed before outputs are analyzed.

Web Appendix D: Complete Results

Study C1: Twin Effects, Human Benchmarks, Discrepancies

Table W5: C1 effects (default effects in percentage points; attraction in log odds; other effects in standardized units of Table 3 of the manuscript). D = twin minus human; interval from 1,000 respondent bootstrap resamples.

Architecture	Effect	Model	Twin	Human	D	D 95% CI
demo	anch_africa	claude-haiku-5.5	0.36	0.78	-0.42	-0.53 to -0.32
demo	anch_africa	gemini-3.7-flash	0.85	0.78	0.07	-0.06 to 0.19
demo	anch_africa	llama-4-maverick	0.85	0.78	0.07	-0.09 to 0.25
demo	anch_africa	gpt-5.6-luna	0.21	0.78	-0.57	-0.65 to -0.49
demo	anch_africa	grok-4.3	0.52	0.78	-0.26	-0.35 to -0.16
demo	anch_redwood	claude-haiku-5.5	0.30	1.69	-1.39	-1.51 to -1.27
demo	anch_redwood	gemini-3.7-flash	1.62	1.69	-0.07	-0.21 to 0.09
demo	anch_redwood	llama-4-maverick	1.39	1.69	-0.30	-0.44 to -0.14
demo	anch_redwood	gpt-5.6-luna	0.16	1.69	-1.53	-1.64 to -1.44
demo	anch_redwood	grok-4.3	0.84	1.69	-0.85	-0.97 to -0.73
demo	attraction	claude-haiku-5.5	-0.10	0.37	-0.47	-0.67 to -0.27
demo	attraction	gemini-3.7-flash	2.07	0.37	1.70	1.41 to 2.05
demo	attraction	llama-4-maverick	-1.86	0.37	-2.23	-2.53 to -1.98
demo	attraction	gpt-5.6-luna	2.02	0.37	1.65	1.40 to 1.92
demo	attraction	grok-4.3	0.76	0.37	0.39	0.13 to 0.69
demo	default_green	claude-haiku-5.5	-31.83	24.67	-56.50	-63.90 to -49.72
demo	default_green	gemini-3.7-flash	1.42	24.67	-23.25	-30.34 to -15.08
demo	default_green	llama-4-maverick	-51.08	24.67	-75.75	-82.53 to -68.97
demo	default_green	gpt-5.6-luna	-33.42	24.67	-58.09	-65.30 to -50.88
demo	default_green	grok-4.3	-4.20	24.67	-28.87	-36.67 to -20.71
demo	default_organ	claude-haiku-5.5	34.34	4.39	29.95	24.41 to 35.34
demo	default_organ	gemini-3.7-flash	2.07	4.39	-2.32	-3.72 to -0.62
demo	default_organ	llama-4-maverick	-69.77	4.39	-74.16	-79.75 to -68.74
demo	default_organ	gpt-5.6-luna	12.44	4.39	8.05	4.48 to 12.24
demo	default_organ	grok-4.3	22.85	4.39	18.46	13.65 to 23.56
demo	fair	claude-haiku-5.5	-0.31	-0.54	0.23	0.05 to 0.38
demo	fair	gemini-3.7-flash	-1.83	-0.54	-1.29	-1.52 to -1.07
demo	fair	llama-4-maverick	-0.45	-0.54	0.09	-0.08 to 0.25
demo	fair	gpt-5.6-luna	-0.45	-0.54	0.09	-0.07 to 0.26
demo	fair	grok-4.3	0.09	-0.54	0.63	0.46 to 0.80
demo	frame	claude-haiku-5.5	0.53	0.76	-0.23	-0.40 to -0.07
demo	frame	gemini-3.7-flash	4.11	0.76	3.35	2.72 to 4.26
demo	frame	llama-4-maverick	2.39	0.76	1.63	1.35 to 1.93
demo	frame	gpt-5.6-luna	-0.19	0.76	-0.95	-1.10 to -0.80
demo	frame	grok-4.3	0.44	0.76	-0.32	-0.41 to -0.22
summary	anch_africa	claude-haiku-5.5	0.61	0.78	-0.17	-0.28 to -0.06
summary	anch_africa	gemini-3.7-flash	1.23	0.78	0.45	0.32 to 0.61
summary	anch_africa	llama-4-maverick	0.85	0.78	0.07	-0.06 to 0.20
summary	anch_africa	gpt-5.6-luna	0.71	0.78	-0.07	-0.17 to 0.04
summary	anch_africa	grok-4.3	0.84	0.78	0.06	-0.05 to 0.18
summary	anch_redwood	claude-haiku-5.5	0.66	1.69	-1.03	-1.15 to -0.88
summary	anch_redwood	gemini-3.7-flash	1.74	1.69	0.05	-0.08 to 0.21
summary	anch_redwood	llama-4-maverick	2.08	1.69	0.39	0.20 to 0.64
summary	anch_redwood	gpt-5.6-luna	0.57	1.69	-1.12	-1.20 to -1.03
summary	anch_redwood	grok-4.3	0.98	1.69	-0.71	-0.80 to -0.61

summary	attraction	claude-haiku-5.5	-0.19	0.37	-0.56	-0.77 to -0.35
summary	attraction	gemini-3.7-flash	1.91	0.37	1.54	1.29 to 1.86
summary	attraction	llama-4-maverick	-0.36	0.37	-0.73	-1.05 to -0.44
summary	attraction	gpt-5.6-luna	0.90	0.37	0.53	0.25 to 0.75
summary	attraction	grok-4.3	-0.29	0.37	-0.66	-0.88 to -0.44
summary	default_green	claude-haiku-5.5	-2.00	24.67	-26.67	-33.83 to -18.72
summary	default_green	gemini-3.7-flash	12.42	24.67	-12.25	-20.28 to -4.86
summary	default_green	llama-4-maverick	-29.30	24.67	-53.97	-62.14 to -45.82
summary	default_green	gpt-5.6-luna	-1.63	24.67	-26.30	-34.17 to -18.43
summary	default_green	grok-4.3	12.31	24.67	-12.36	-20.55 to -4.76
summary	default_organ	claude-haiku-5.5	31.37	4.39	26.98	21.10 to 33.18
summary	default_organ	gemini-3.7-flash	6.90	4.39	2.51	-0.49 to 5.36
summary	default_organ	llama-4-maverick	-27.28	4.39	-31.67	-39.42 to -23.59
summary	default_organ	gpt-5.6-luna	21.92	4.39	17.53	12.65 to 22.81
summary	default_organ	grok-4.3	32.38	4.39	27.99	22.00 to 34.46
summary	fair	claude-haiku-5.5	-0.26	-0.54	0.28	0.13 to 0.44
summary	fair	gemini-3.7-flash	-1.86	-0.54	-1.32	-1.57 to -1.11
summary	fair	llama-4-maverick	0.23	-0.54	0.77	0.59 to 0.95
summary	fair	gpt-5.6-luna	-0.29	-0.54	0.25	0.09 to 0.42
summary	fair	grok-4.3	0.40	-0.54	0.94	0.77 to 1.10
summary	frame	claude-haiku-5.5	0.38	0.76	-0.38	-0.54 to -0.21
summary	frame	gemini-3.7-flash	5.95	0.76	5.19	4.12 to 7.07
summary	frame	llama-4-maverick	1.77	0.76	1.01	0.75 to 1.29
summary	frame	gpt-5.6-luna	-0.15	0.76	-0.91	-1.07 to -0.74
summary	frame	grok-4.3	0.52	0.76	-0.24	-0.39 to -0.08
theory	anch_africa	claude-haiku-5.5	0.45	0.78	-0.33	-0.42 to -0.21
theory	anch_africa	gemini-3.7-flash	1.75	0.78	0.97	0.79 to 1.17
theory	anch_africa	llama-4-maverick	1.27	0.78	0.49	0.39 to 0.61
theory	anch_africa	gpt-5.6-luna	0.80	0.78	0.02	-0.08 to 0.14
theory	anch_africa	grok-4.3	0.90	0.78	0.12	0.00 to 0.24
theory	anch_redwood	claude-haiku-5.5	0.42	1.69	-1.27	-1.42 to -1.11
theory	anch_redwood	gemini-3.7-flash	3.00	1.69	1.31	1.07 to 1.60
theory	anch_redwood	llama-4-maverick	5.61	1.69	3.92	3.14 to 4.88
theory	anch_redwood	gpt-5.6-luna	0.84	1.69	-0.85	-0.93 to -0.75
theory	anch_redwood	grok-4.3	0.99	1.69	-0.70	-0.79 to -0.60
theory	attraction	claude-haiku-5.5	0.40	0.37	0.03	-0.20 to 0.23
theory	attraction	gemini-3.7-flash	3.03	0.37	2.66	2.40 to 2.98
theory	attraction	llama-4-maverick	-0.24	0.37	-0.61	-0.87 to -0.36
theory	attraction	gpt-5.6-luna	2.60	0.37	2.23	2.03 to 2.45
theory	attraction	grok-4.3	2.15	0.37	1.78	1.52 to 2.04
theory	default_green	claude-haiku-5.5	-14.90	24.67	-39.57	-47.13 to -31.93
theory	default_green	gemini-3.7-flash	6.71	24.67	-17.96	-25.51 to -9.19
theory	default_green	llama-4-maverick	-72.08	24.67	-96.75	-102.31 to -91.11
theory	default_green	gpt-5.6-luna	39.29	24.67	14.62	6.97 to 21.59
theory	default_green	grok-4.3	20.43	24.67	-4.24	-11.90 to 3.66
theory	default_organ	claude-haiku-5.5	48.20	4.39	43.81	37.26 to 49.58
theory	default_organ	gemini-3.7-flash	5.52	4.39	1.13	-1.34 to 3.94
theory	default_organ	llama-4-maverick	-61.87	4.39	-66.26	-72.46 to -60.15
theory	default_organ	gpt-5.6-luna	62.07	4.39	57.68	52.41 to 63.15
theory	default_organ	grok-4.3	40.29	4.39	35.90	29.88 to 41.94
theory	fair	claude-haiku-5.5	-0.20	-0.54	0.34	0.19 to 0.51
theory	fair	gemini-3.7-flash	-1.64	-0.54	-1.10	-1.35 to -0.89
theory	fair	llama-4-maverick	-0.07	-0.54	0.47	0.30 to 0.62
theory	fair	gpt-5.6-luna	0.12	-0.54	0.66	0.50 to 0.81
theory	fair	grok-4.3	-0.02	-0.54	0.52	0.36 to 0.69
theory	frame	claude-haiku-5.5	0.49	0.76	-0.27	-0.42 to -0.12

theory	frame	gemini-3.7-flash	8.54	0.76	7.78	5.98 to 11.69
theory	frame	llama-4-maverick	2.34	0.76	1.58	1.32 to 1.90
theory	frame	gpt-5.6-luna	-0.22	0.76	-0.98	-1.15 to -0.82
theory	frame	grok-4.3	1.44	0.76	0.68	0.53 to 0.85

Table W6: C1 registered propositions and robustness checks.

Test	Pooled	95% CI	Models in direction	p (boot)	q (BH)	Verdict
P1 framing D>0 (summary)	0.936	0.718 to 1.325	2	0.001	0.0010	not supported
P1 anchoring D>0 (summary)	-0.207	-0.248 to -0.156	2	0.001	0.0010	not supported
P2a organ default D>0 (summary)	8.667	6.094 to 11.172	4	0.001	0.0010	supported
P2b green default D<=0 (summary)	-26.308	-29.821 to -22.846	5	0.001	0.0010	supported
P3 defaults D_green-D_organ<0 (summary)	-34.976	-39.288 to -30.937	5	0.001	0.0010	supported
P1 framing D>0 (demo, robustness)	0.697	0.557 to 0.889	2	0.001		not supported
P1 anchoring D>0 (demo, robustness)	-0.525	-0.566 to -0.481	0	0.001		not supported
P2a organ default D>0 (demo, robustness)	-4.007	-6.030 to -2.089	3	0.001		not supported
P2b green default D<=0 (demo, robustness)	-48.490	-51.796 to -44.965	5	0.001		supported
P3 defaults (demo, robustness)	-44.483	-48.082 to -40.823	5	0.001		supported
P6 theory minus demo, canonical paradigms	0.713	0.577 to 0.971	5	0.001	0.0010	supported

Study H6c: Reasoning Grid Cells

Table W7: H6c cell estimates (100 respondents per cell).

Model	Effort	Temp	n	Framing d	Attraction	Anch. Africa	Anch. redwood	Bias index	Error	Fabr. A	Fabr. B
claude-haiku-5.5	none	0.0	100	0.44	0.17	0.73	1.11	0.61	0.149	0.005	0.002
claude-haiku-5.5	none	0.7	100	0.39	0.16	0.74	1.10	0.58	0.151	0.005	0.003
claude-haiku-5.5	none	1.2	100	0.42	-0.11	0.60	1.06	0.32	0.156	0.006	0.002
claude-haiku-5.5	low	0.0	100	0.29	-0.27	0.38	0.51	0.01	0.012	0.000	0.000
claude-	low	0.7	100	0.20	-0.53	0.35	0.22	-0.30	0.010	0.000	0.000

haiku-5.5											
claude-haiku-5.5	low	1.2	100	0.06	-0.69	0.27	0.30	-0.51	0.011	0.000	0.000
claude-haiku-5.5	high	0.0	100	0.40	-0.21	0.53	0.94	0.19	0.002	0.000	0.000
claude-haiku-5.5	high	0.7	100	0.58	-0.39	0.59	0.67	0.09	0.003	0.000	0.000
claude-haiku-5.5	high	1.2	100	0.53	-0.51	0.59	0.82	-0.02	0.002	0.000	0.000
gemini-3.7-flash	low	0.0	100	3.31	2.20	0.73	1.44	3.73	0.000	0.000	0.000
gemini-3.7-flash	low	0.7	100	2.92	2.49	0.75	1.50	3.83	0.000	0.000	0.000
gemini-3.7-flash	low	1.2	100	3.23	2.06	0.76	1.54	3.58	0.000	0.000	0.000
gemini-3.7-flash	medium	0.0	100	6.21	1.50	1.11	1.78	4.49	0.000	0.000	0.000
gemini-3.7-flash	medium	0.7	100	6.21	1.91	0.92	1.81	4.82	0.000	0.000	0.000
gemini-3.7-flash	medium	1.2	100	5.06	1.88	0.98	1.59	4.28	0.000	0.000	0.000
gemini-3.7-flash	high	0.0	100	4.26	1.75	1.24	1.92	3.89	0.000	0.000	0.000
gemini-3.7-flash	high	0.7	100	6.71	1.51	1.12	1.77	4.71	0.000	0.000	0.000
gemini-3.7-flash	high	1.2	100	6.00	1.97	1.09	1.66	4.80	0.000	0.000	0.000
gpt-5.6-luna	none	0.0	100	-0.35	2.19	0.00	0.13	1.83	0.049	0.001	0.000
gpt-5.6-luna	none	0.7	100	-0.16	1.28	0.00	-0.18	1.07	0.047	0.000	0.000
gpt-5.6-luna	none	1.2	100	-0.26	2.27	0.00	-0.05	1.93	0.048	0.001	0.000
gpt-5.6-luna	low	0.0	100	-0.30	1.84	0.00	0.00	1.52	0.014	0.000	0.003
gpt-5.6-luna	low	0.7	100	-0.26	2.23	0.27	0.20	1.97	0.014	0.000	0.002
gpt-5.6-luna	low	1.2	100	-0.29	1.80	0.19	0.01	1.54	0.017	0.000	0.001
gpt-5.6-luna	high	0.0	100	-0.57	2.28	0.20	0.34	1.88	0.000	0.000	0.001
gpt-5.6-luna	high	0.7	100	0.07	2.50	0.32	0.44	2.39	0.000	0.000	0.000
gpt-5.6-luna	high	1.2	100	-0.56	2.21	0.00	0.26	1.77	0.001	0.000	0.001
grok-4.3	none	0.0	100	0.50	-0.67	0.00	0.32	-0.35	0.003	0.000	0.000
grok-4.3	none	0.7	100	0.56	-0.69	0.43	0.51	-0.23	0.006	0.000	0.000
grok-4.3	none	1.2	100	0.28	-0.57	0.34	0.72	-0.25	0.009	0.000	0.000
grok-4.3	low	0.0	100	0.39	0.77	0.47	0.59	1.02	0.000	0.000	0.000
grok-4.3	low	0.7	100	0.38	0.35	0.55	0.72	0.67	0.000	0.000	0.000
grok-4.3	low	1.2	100	0.62	0.61	0.74	0.69	1.05	0.000	0.000	0.000
grok-4.3	high	0.0	100	0.72	0.83	0.67	1.11	1.32	0.000	0.000	0.000
grok-4.3	high	0.7	100	0.68	0.51	0.62	1.21	1.01	0.000	0.000	0.000
grok-4.3	high	1.2	100	0.63	0.67	0.83	1.18	1.17	0.000	0.000	0.000

Table W8: H6c registered tests.

Test	Coefficient	95% interval	Models in direction	Verdict
P4 level -> bias_idx (predicted negative)	0.272	0.082 to 0.462	1/4	not supported
P4 level -> frame (predicted negative)	0.340	0.034 to 0.645	1/4	not supported
P4 level -> attraction (predicted negative)	0.096	-0.097 to 0.288	2/4	not supported

P4 level -> err (predicted negative)	-0.025	-0.039 to -0.011	3/4	supported
P4 level -> fabA (predicted negative)	-0.001	-0.001 to -0.000	2/4	not supported
P4 level -> fabB (predicted negative)	-0.000	-0.001 to 0.000	1/4	not supported
P5 within-model Spearman(bias_idx, error) (predicted positive)	-0.192	-0.408 to 0.412	1/4 positive	not supported

Study C2: Field Test

Table W9: C2 per-model summaries by arm and test class.

Model	Arm	Cue-contrast	Tests	Significant tests	Agreement	Chance	Mean rho	Raw slope	Disatt. slope
claude-haiku-5.5	nopersona	False	166	65	0.415	0.244	0.034	0.057	0.082
claude-haiku-5.5	nopersona	True	833	463	0.287	0.216	0.122	0.108	0.159
claude-haiku-5.5	persona	False	166	65	0.200	0.244	-0.027	-0.010	-0.015
claude-haiku-5.5	persona	True	832	462	0.214	0.216	-0.021	-0.028	-0.041
gemini-3.7-flash	nopersona	False	166	65	0.323	0.244	0.030	0.104	0.149
gemini-3.7-flash	nopersona	True	833	463	0.322	0.216	0.187	0.236	0.346
gemini-3.7-flash	persona	False	166	65	0.292	0.244	0.006	-0.002	-0.003
gemini-3.7-flash	persona	True	833	463	0.212	0.216	-0.006	-0.050	-0.073
llama-4-maverick	nopersona	False	166	65	0.277	0.244	0.103	0.213	0.304
llama-4-maverick	nopersona	True	833	463	0.283	0.216	0.111	0.206	0.303
llama-4-maverick	persona	False	166	65	0.292	0.244	0.051	0.079	0.113
llama-4-maverick	persona	True	833	463	0.210	0.216	0.002	-0.089	-0.131
gpt-5.6-luna	nopersona	False	166	65	0.354	0.244	0.042	0.063	0.091
gpt-5.6-luna	nopersona	True	833	463	0.320	0.216	0.129	0.146	0.214
gpt-5.6-luna	persona	False	166	65	0.338	0.244	0.011	0.026	0.037
gpt-5.6-luna	persona	True	833	463	0.251	0.216	0.043	0.014	0.021
grok-4.3	nopersona	False	166	65	0.385	0.244	0.049	0.120	0.172
grok-4.3	nopersona	True	833	463	0.320	0.216	0.127	0.179	0.263
grok-4.3	persona	False	166	65	0.277	0.244	-0.006	-0.023	-0.033
grok-4.3	persona	True	833	463	0.190	0.216	-0.034	-0.072	-0.105

Table W10: C2 registered verdicts with per-model values.

Test	Pooled	95% CI	Models	q (BH)	Verdict	claude-haiku-5.5	gemini-3.7-flash	llama-4-maverick	gpt-5.6-luna	grok-4.3
P7a cue minus non-cue excess agreement (no-persona)	-0.016	-0.100 to 0.065	2/5	0.6600	not supported	-0.100	0.027	0.034	-0.006	-0.037
P7b disattenuated slope in	0.257	0.199 to 0.314	0/5	0.0013	not supported	0.159	0.346	0.303	0.214	0.263

cue tests (no-persona), null = 1										
P7c no-persona minus persona excess agreement (all tests)	0.089	0.065 to 0.116	5/5	0.0013	supported	0.091	0.100	0.062	0.062	0.127
P7d no-persona excess agreement (all tests)	0.093	0.066 to 0.121	5/5	0.0013	supported	0.084	0.103	0.063	0.105	0.108
robustness: P7a persona arm	-0.036	-0.125 to 0.050	1/5		not supported	0.043	-0.052	-0.054	-0.059	-0.058
robustness: P7b persona arm, null = 1	-0.066	-0.131 to 0.000	0/5		not supported	-0.041	-0.073	-0.131	0.021	-0.105

Cue-contrast rule: digit present; question mark; word from the lexicon {worst, terrible, horrible, tragic, shocking, disgusting, awful, scary, never, stop, dangerous, fail(s/ed), lose/lost/losing, dead, death, kill(s), hate, angry, outrage, crisis, nightmare, ruin(ed)}; second-person pronoun (you, your, yours). A test is cue-contrast if the headline arms disagree on at least one of the four features.

Study S: Validity Levels and Stability

Table W11: Validity level per model, template, and effect (2 = magnitude-valid, 1 = direction-valid, 0 = invalid).

Model	Template	anch_africa	anch_redwood	attraction	default_green	default_organ	fair	frame
claude-haiku-4.5	A	1	1	1	0	0	0	1
claude-haiku-4.5	B	1	1	2	0	0	0	1
claude-haiku-5.5	A	1	1	2	0	0	2	2
claude-haiku-5.5	B	1	1	0	0	0	2	1
claude-sonnet-5	A	1	1	1	1	0	0	1
claude-sonnet-5	B	1	1	0	1	0	0	1
claude-sonnet-5.5	A	1	2	1	2	0	1	1
claude-sonnet-5.5	B	1	1	1	1	0	2	1
gemini-3.5-flash	A	2	1	1	0	0	1	1
gemini-3.5-flash	B	1	1	1	0	0	1	1
gemini-3.6-flash	A	1	1	1	0	0	2	1
gemini-3.6-flash	B	2	1	1	0	0	2	1
gemini-3.7-flash	A	1	1	1	0	2	1	1
gemini-3.7-	B	1	2	1	0	2	1	1

flash								
gpt-5.2	A	1	1	1	0	2	1	0
gpt-5.2	B	1	1	1	0	2	1	0
gpt-5.4	A	1	1	1	0	0	1	1
gpt-5.4	B	1	1	1	0	0	1	1
gpt-5.5	A	1	1	1	0	0	1	1
gpt-5.5	B	1	1	1	0	0	1	1
grok-4.3	A	1	1	0	0	0	0	1
grok-4.3	B	2	1	0	0	0	0	1
grok-4.5	A	1	1	1	0	0	2	1
grok-4.5	B	1	1	2	0	0	2	1
grok-4.7	A	1	2	1	0	0	1	1
grok-4.7	B	1	2	1	0	0	1	2

Table W12: Twin effect estimates in Study S (human benchmarks in Table 3).

Model	Template	anch_africa	anch_redwood	attraction	default_green	default_organ	fair	frame
claude-haiku-4.5	A	0.54	1.45	0.86	5.39	29.35	1.03	0.53
claude-haiku-4.5	B	0.65	1.38	0.19	5.12	30.78	0.71	0.28
claude-haiku-5.5	A	1.22	0.85	0.34	-36.11	28.77	-0.33	0.78
claude-haiku-5.5	B	0.50	0.64	-0.07	-16.34	35.49	-0.33	0.29
claude-sonnet-5	A	3.01	0.87	0.66	54.35	41.33	1.89	0.36
claude-sonnet-5	B	2.49	0.89	-0.19	54.83	52.68	0.89	0.36
claude-sonnet-5.5	A	0.98	1.60	1.76	19.00	40.71	-0.89	4.54
claude-sonnet-5.5	B	1.12	1.90	1.84	32.97	66.37	-0.56	1.70
gemini-3.5-flash	A	0.77	0.83	1.60	-5.31	24.87	-0.99	1.65
gemini-3.5-flash	B	0.65	0.81	1.01	-1.89	19.03	-0.89	1.23
gemini-3.6-flash	A	0.56	1.24	2.22	-5.31	26.55	-0.70	3.29
gemini-3.6-flash	B	0.79	0.98	2.10	-7.00	20.71	-0.44	2.32
gemini-3.7-flash	A	1.02	1.51	2.22	0.04	5.31	-1.78	3.88
gemini-3.7-flash	B	1.20	1.69	2.45	4.39	7.70	-1.89	5.15
gpt-5.2	A	0.38	0.84	2.11	-10.35	6.91	-1.25	0.14
gpt-5.2	B	0.57	0.70	1.60	-8.66	7.26	-1.11	-0.40
gpt-5.4	A	0.61	1.12	2.40	-9.42	33.46	-1.76	1.75
gpt-5.4	B	0.66	1.03	2.60	-10.99	23.46	-1.69	1.43
gpt-5.5	A	0.65	0.80	1.30	-5.88	16.81	-1.61	0.93
gpt-5.5	B	0.58	0.96	1.47	-4.79	15.04	-1.51	0.46
grok-4.3	A	0.93	1.11	-0.04	-1.85	32.31	0.14	0.41
grok-4.3	B	0.70	1.07	-0.17	-7.13	32.57	0.59	0.40
grok-4.5	A	1.35	1.20	0.78	-4.03	38.06	-0.72	2.25
grok-4.5	B	1.13	1.28	0.61	-0.60	50.27	-0.58	1.99
grok-4.7	A	1.00	1.70	0.87	11.47	25.05	-1.23	0.88
grok-4.7	B	1.12	1.69	0.99	7.93	38.24	-1.32	0.81

{('anthropic/claude-haiku-4.5', 'B'): '13/200 excl', ('anthropic/claude-haiku-4.5', 'A'): '97/200 excl'}
Stability statistics: Template stability: classification agreement A vs B = 0.86 (kappa 0.75); correlation of z-scored discrepancies = 0.87 (n = 91 model-effect pairs)
model 0.25
effect 0.00
template 0.00

model x effect 0.68
model x template 0.01
effect x template 0.01
residual (3-way) 0.05
model main effect splits into family 0.78 and generation-within-family 0.22

Exploratory Studies E1 and E2

Table W13: E1 twin effects by model and architecture (template A; defaults in points, attraction in log odds, fairness d). Human row first.

source	default_green_pp	default_organ_pp	attraction_logOR	fair_d	n_rows
human	24.67	4.39	0.37	-0.54	
anthropic/claude-haiku-5.5|demo	-14.5	45.84	0.56	-0.91	6120.0
anthropic/claude-haiku-5.5|summary	-2.33	26.56	0.48	-0.57	6120.0
anthropic/claude-haiku-5.5|theory	-2.33	49.16	1.14	-1.57	3060.0
google/gemini-3.7-flash|demo	15.17	6.48	1.49	-1.65	6120.0
google/gemini-3.7-flash|summary	18.5	10.96	1.55	-1.92	6108.0
google/gemini-3.7-flash|theory	25.33	24.25	2.57	-1.98	3060.0
meta-llama/llama-4-maverick|demo	-23.11	1.26	1.68	0.52	5559.0
meta-llama/llama-4-maverick|summary	-21.05	7.86	1.46	0.4	5534.0
meta-llama/llama-4-maverick|theory	-28.65	18.11	2.09	-0.48	2782.0
openai/gpt-5.6-luna|demo	-2.17	7.8	2.77	0.12	6120.0
openai/gpt-5.6-luna|summary	-12.17	6.62	2.73	0.15	6120.0
openai/gpt-5.6-luna|theory	38.33	36.87	2.6	-0.28	3060.0
x-ai/grok-4.3|demo	4.67	29.21	1.23	0.29	6120.0
x-ai/grok-4.3|summary	1.17	23.54	0.65	0.76	6120.0
x-ai/grok-4.3|theory	24.67	47.16	2.58	-0.25	3060.0

(HC1 SEs), cells as units
bias_idx n=36 level: +0.541 (se 0.114) temp: -0.028 (se 0.180)
green_pp n=36 level: -0.804 (se 4.513) temp: -0.073 (se 6.384)
attraction n=36 level: +0.468 (se 0.066) temp: -0.006 (se 0.114)
fair_d n=36 level: -0.212 (se 0.075) temp: +0.035 (se 0.123)
hall n=36 level: -0.016 (se 0.004) temp: +0.000 (se 0.007)
fab n=36 level: -0.001 (se 0.000) temp: -0.000 (se 0.001)
err n=36 level: -0.025 (se 0.007) temp: +0.000 (se 0.011)

Cell-level Spearman, bias_idx vs hallucination: pooled rho=-0.53 (n=36); within-model rho=-0.79

The exploratory H6 pilot pointed to a positive relation between reasoning level and bias and a negative within-cell correlation of bias with error; the confirmatory grid H6c replicates the first and the second at smaller size (Section 4.3 of the manuscript).

Web Appendix E: Data Quality, Exclusions, and Known Issues

• Invalid responses (not parseable JSON or out-of-range option) are excluded and counted. In C1, the Llama 4 Maverick persona-summary cell excluded 85 of 600 responses (14%), above the registered 10% flag; two Llama theory-explicit and four GPT-5.6 luna persona-summary responses were also excluded; all other C1 cells had no exclusions. In Study S, Claude Haiku 4.5 with template A excluded 97 of 200 responses (49%) and with template B 13 of 200; its template A estimates should not be interpreted.

• E1 lacks 441 planned calls because of repeated upstream rate-limit failures for Llama 4 Maverick. Retries were attempted up to five times per call and a second pass filled most failed calls.

• The raw condition tag (EmbeddedData "target") in the targeting-fairness mega-study file is inverted relative to the survey text shown to participants. All analyses code conditions from the displayed text.

• An indexing error in our Benjamini–Hochberg routine was found after the C2 evaluation (a p-value of 0.66 received q = 0.001). It was corrected and C1 and C2 were recomputed; no verdict changed. The corrected code and results are in the repository, and the history documents the sequence.

• The Upworthy archive version used (December 2020 files) has no randomization-risk flag; tests created from 2013-06-25 to 2014-01-10 were excluded by date. The archive is public and could be partly present in model training data.

• The aggregator selects the serving provider; provider identity, quantization, and routing were not controlled or recorded. Reasoning-effort semantics differ across providers.

• Human benchmark intervals for loss framing and the two anchoring items come from the Twin-2K-500 repeat wave (wave 4); the interval for targeting fairness is an approximate Cohen's d interval computed from the group sizes.

Web Appendix F: Code, Data, and Disclosures

Repository contents (see README in the code package): pilot/ (runners and analysis scripts), benchmarks/ (human-benchmark computation), prereg/ (registrations, amendment, working document), results/ (summary tables), paper_src/ (manuscript source). Raw model outputs (JSON lines, about 300 MB uncompressed, 18 MB compressed) are provided as a separate archive. Human data are public and are not redistributed: Twin-2K-500 (CC BY 4.0, huggingface.co/datasets/LLM-Digital-Twin/Twin-2K-500), the Twin-2K-500 mega-study (Apache-2.0, huggingface.co/datasets/LLM-Digital-Twin/Twin-2K-500-Mega-Study), and the Upworthy Research Archive (OSF project jd64p).

Reproduction: (1) download the public datasets; (2) python3 benchmarks/compute_human_benchmarks.py <data_dir> <data_dir> to recompute human benchmarks; (3) run the study runners in pilot/ (they need an API key for an LLM aggregator and write JSON-lines files); (4) run the matching analyze_*.py scripts; (5) node paper_src/make.js to rebuild the manuscript. Analysis scripts for confirmatory studies were committed before their outputs were analyzed (see commit history).

Software: Python 3.13 (pandas, numpy, scipy, requests, pyarrow, python-docx), Node.js with docx. Random seeds: 2026 (respondent draws, C2 test draw, bootstrap), 11 (C2 development draw), 2027 (Study S respondent draw).

Human subjects: no new human-subject data were collected; all human data are public secondary data. Institutional determination of exemption: [to be completed by the authors].

Disclosure of AI assistance: an AI model (Anthropic Claude) assisted with study design discussion, writing the simulation and analysis code, running analyses, and drafting the manuscript and this appendix. The authors are responsible for the content and should verify every number against the repository before submission: [to be confirmed by the authors in line with journal policy].